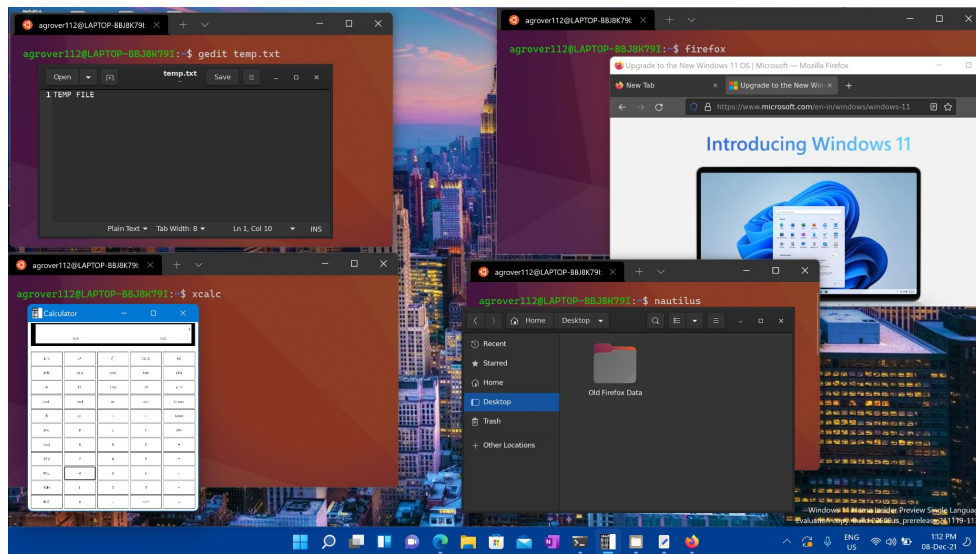


LINUX



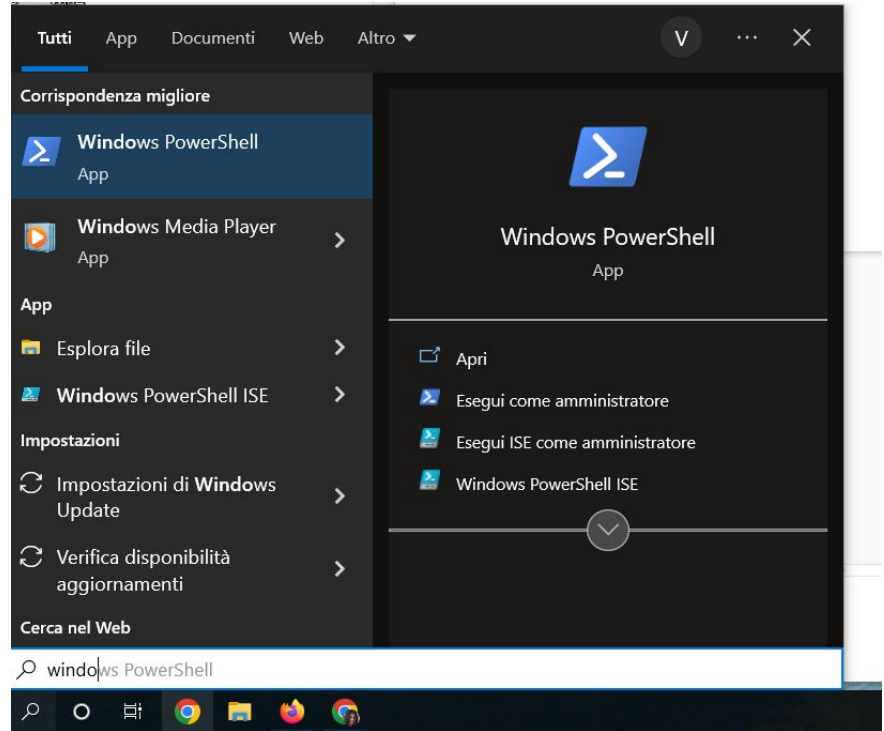
Install WSL

Windows Subsystem for Linux (WSL) allows you to install a complete Ubuntu terminal environment in minutes on your Windows machine, allowing you to develop cross-platform applications without leaving Windows.



Install WSL

1. Search for Windows PowerShell in your Windows search bar, then select Run as administrator.



Install WSL

1. Search for Windows PowerShell in your Windows search bar, then select **Run as administrator**.

```
vlukaj@DESKTOP-P2RPBKF: /mnt/c/Windows/system32
Welcome to Ubuntu 20.04.2 LTS (GNU/Linux 4.4.0-18362-Microsoft x86_64)

Documentation:  https://help.ubuntu.com
Management:    https://landscape.canonical.com
Support:       https://ubuntu.com/advantage

System information as of Wed Nov  9 09:35:19 CET 2022

System load:  0.52      Users logged in:  0
Usage of /home: unknown IPv4 address for eth0: 172.17.2.68
Memory usage: 26%      IPv4 address for eth1: 192.168.56.1
Swap usage:   0%        IPv4 address for eth2: 192.168.99.1

=> There were exceptions while processing one or more plugins. See
   /home/vlukaj/.landscape/sysinfo.log for more information.

0 updates can be installed immediately.
5 of these updates are security updates.
To see these additional updates run: apt list --upgradable

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

This message is shown once a day. To disable it please create the
/home/vlukaj/.hushlogin file.
vlukaj@DESKTOP-P2RPBKF: /mnt/c/Windows/system32$
```

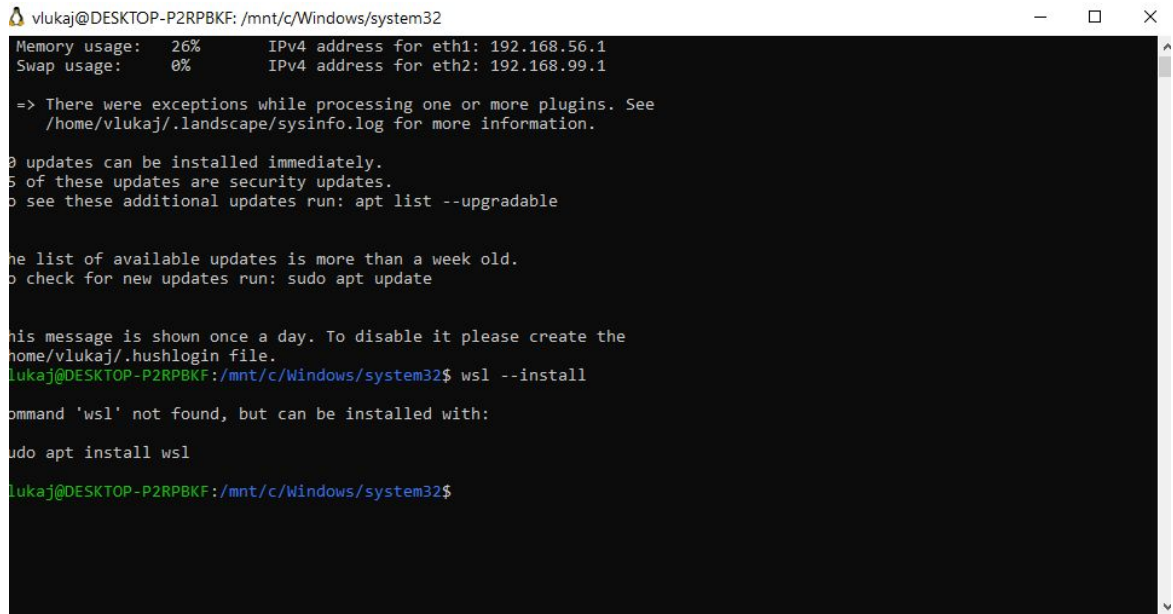
Install WSL

At the command prompt type:

wsl --install

And wait for the process to complete.

For WSL to be properly activated, you will now need to restart your computer.

A terminal window titled 'vlukaj@DESKTOP-P2RPBKF: /mnt/c/Windows/system32' with standard window controls. It displays system information (Memory usage: 26%, Swap usage: 0%, IPv4 addresses for eth1 and eth2), a warning about plugin exceptions, update availability (3 updates, 5 security updates), and instructions to run 'apt list --upgradable' and 'sudo apt update'. It then shows the command 'wsl --install' being entered, followed by a message that 'wsl' is not found but can be installed with 'sudo apt install wsl'. The prompt returns to 'vlukaj@DESKTOP-P2RPBKF: /mnt/c/Windows/system32\$'.

```
vlukaj@DESKTOP-P2RPBKF: /mnt/c/Windows/system32
Memory usage: 26%      IPv4 address for eth1: 192.168.56.1
Swap usage: 0%        IPv4 address for eth2: 192.168.99.1
=> There were exceptions while processing one or more plugins. See
    /home/vlukaj/.landscape/sysinfo.log for more information.

3 updates can be installed immediately.
5 of these updates are security updates.
to see these additional updates run: apt list --upgradable

The list of available updates is more than a week old.
to check for new updates run: sudo apt update

This message is shown once a day. To disable it please create the
file /home/vlukaj/.hushlogin.
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Windows/system32$ wsl --install
Command 'wsl' not found, but can be installed with:
sudo apt install wsl
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Windows/system32$
```

Install WSL

At the command prompt
type:

```
wsl --install
```

And wait for the process to
complete.

For WSL to be properly
activated, you will now need
to restart your computer.

```
sudo apt install wsl
```

```
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Windows/system32$ sudo apt install wsl
[sudo] password for vlukaj:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libxml2-utils xsltproc
The following NEW packages will be installed:
  libxml2-utils wsl xsltproc
0 upgraded, 3 newly installed, 0 to remove and 63 not upgraded.
Need to get 71.4 kB of archives.
After this operation, 476 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu focal/main amd64 libxml2-utils amd64 2.9.10+dfsg-5 [36.9 kB]
Get:2 http://archive.ubuntu.com/ubuntu focal/universe amd64 wsl all 0.2.1-1 [20.1 kB]
Get:3 http://archive.ubuntu.com/ubuntu focal/main amd64 xsltproc amd64 1.1.34-4 [14.3 kB]
Fetched 71.4 kB in 0s (144 kB/s)
```

Install WSL

At the command prompt type:

wsl --install

And wait for the process to complete.

For WSL to be properly activated, you will now need to restart your computer.

```
vlukaj@DESKTOP-P2RPBKF: /mnt/c/Windows/system32
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Windows/system32$ wsl --install
Unknown command: --install
WSL
Isman Shell commandLine, version 0.2.1
USAGE: wsl COMMAND [PARAMS...]

COMMANDS:
Identify - WS-Identify
Enum     - WS-Enumerate
Get      - WS-Get
Put      - WS-Put
Invoke   - WS-Invoke
Clean    - Delete all files generated by this tool set
Cred     - Create or display credential file
Cert     - Get server certificate (saved to <IPADDRESS>.crt)

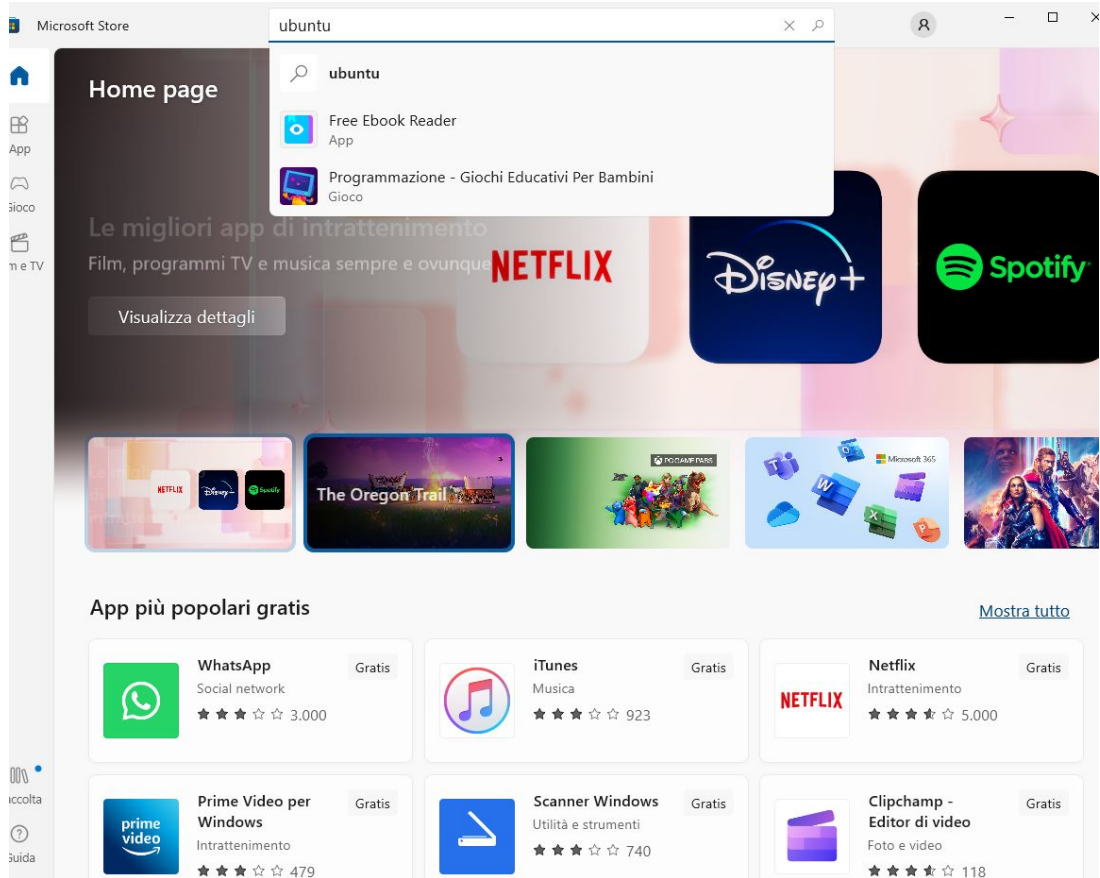
PARAMS specification is specific to a COMMAND.

Output will be saved to ./response.xml. If you want to run parallel
executions in the same directory, define RTFILEPREFIX in the environment.
Doing so may significantly increase files generated.

Requires: curl, xmllint, GNU core utilities.
Optional: xsltproc for output formatting, gpg for encrypted credential.
Optional: wget as alternate for curl when not available.
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Windows/system32$ _
```

Download Ubuntu

WSL supports a variety of Linux distributions, including the [latest Ubuntu release](#), Ubuntu 20.04 LTS and Ubuntu 18.04 LTS. You can find them by opening the Microsoft store app and searching for Ubuntu.



Download Ubuntu

WSL supports a variety of Linux distributions, including the [latest Ubuntu release](#), Ubuntu 20.04 LTS and Ubuntu 18.04 LTS. You can find them by opening the Microsoft store app and searching for Ubuntu.

"ubuntu"

Filtri

Tutti i dipartimenti App Giochi Film Programmi TV



Ubuntu
App • Strumenti di sviluppo
Canonical Group Limited

Ottieni

4,3 ★
Media

20
Classificazioni

3 PEGI
Fascia d'età: PEGI, PEGI 3

Install a complete Ubuntu terminal environment in minutes with Windows Subsystem for Linux (WSL). Develop cross-platform applications, improve your data science or web development...



Ubuntu 20.04.5 LTS
App • Strumenti di sv...
★★★★☆ 14



Ubuntu 22.04.1 LTS
App • Strumenti di sv...
★★★★☆ 19



Ubuntu (Preview)
App • Strumenti di sv...
★★★★★ 1



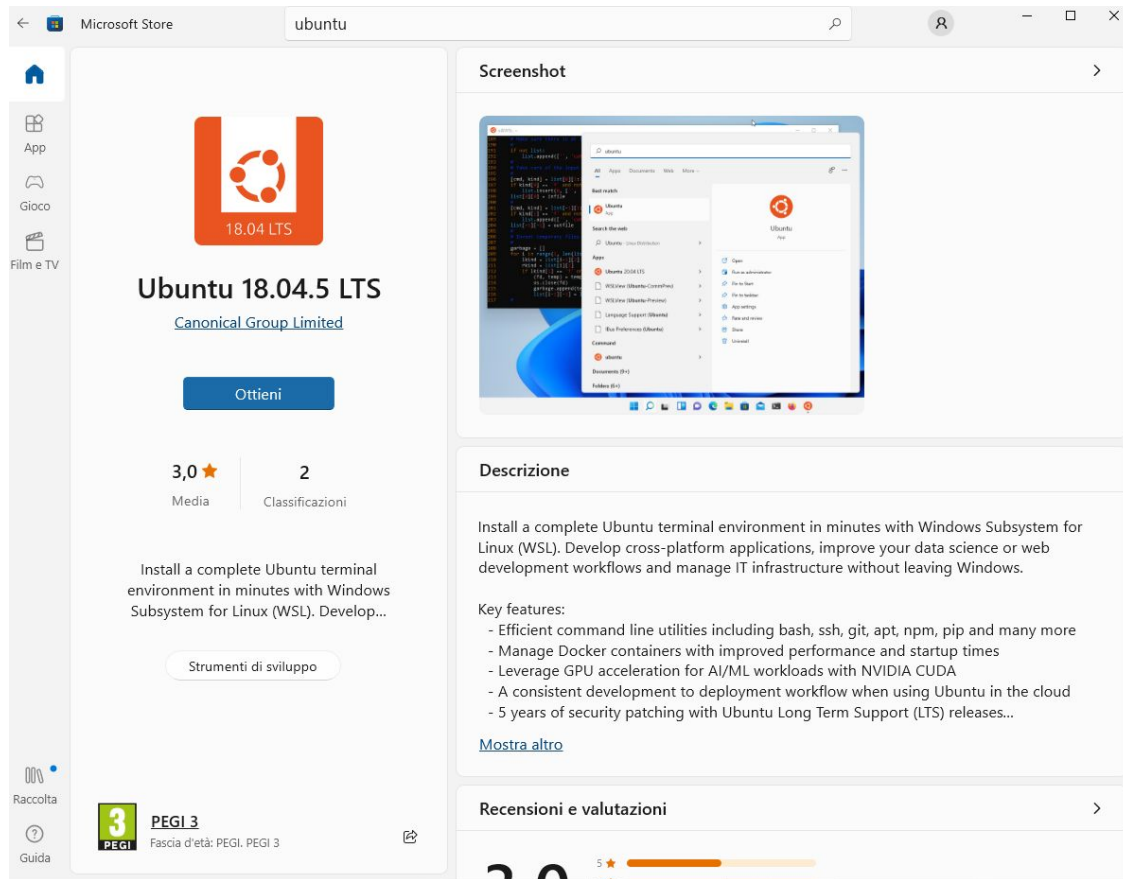
Ubuntu 18.04.5 LTS
App • Strumenti di sv...
★★★★☆ 2



Hoops Africa: Ubuntu Matters
Film • 2018

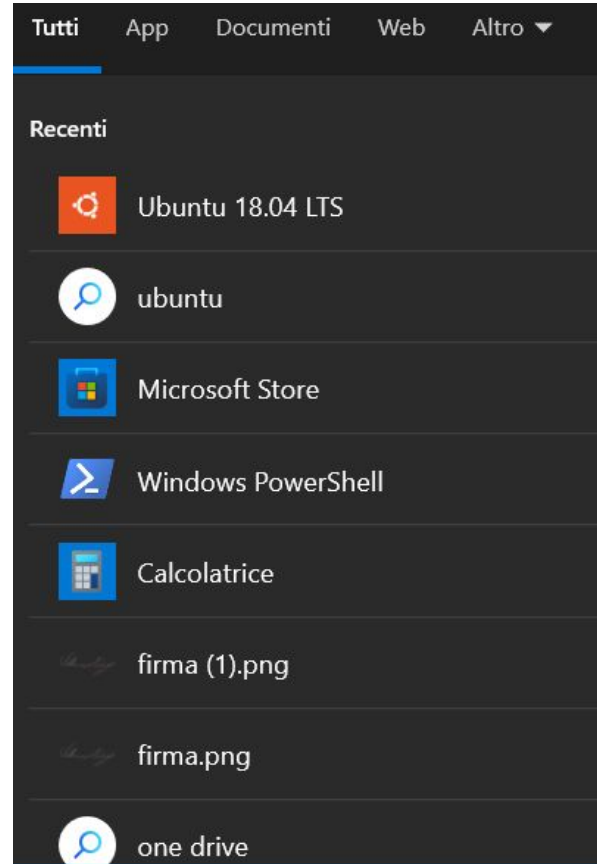
Download Ubuntu

WSL supports a variety of Linux distributions, including the [latest Ubuntu release](#), Ubuntu 20.04 LTS and Ubuntu 18.04 LTS. You can find them by opening the Microsoft store app and searching for Ubuntu.



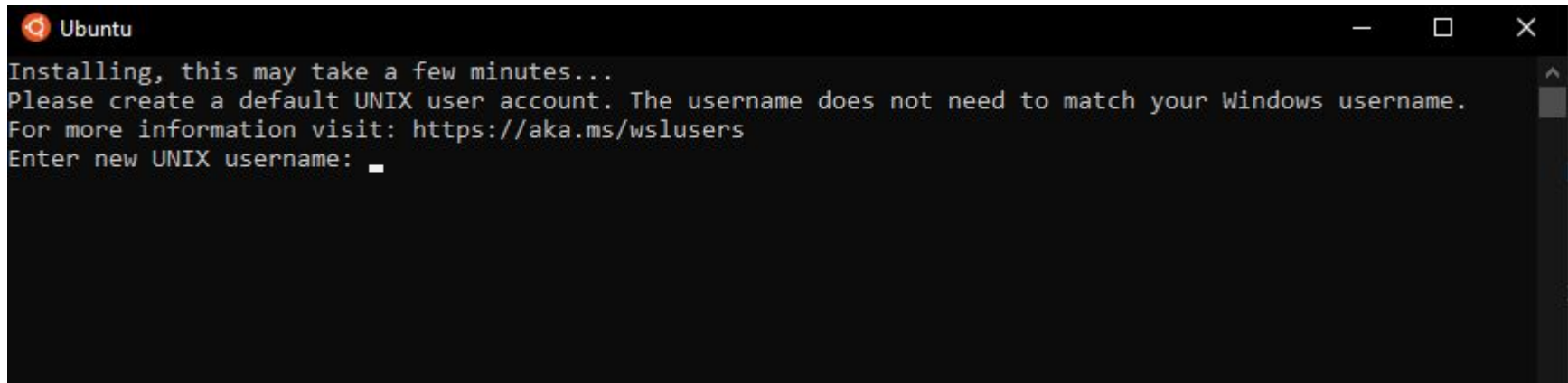
Configure Ubuntu

Once installed, you can either launch the application directly from the store or search for Ubuntu in your Windows search bar.



Configure Ubuntu

Once Ubuntu has finished its initial setup you will need to create a username and password (this does not need to match your Windows user credentials).

A screenshot of a terminal window titled 'Ubuntu' with standard window controls (minimize, maximize, close). The terminal text reads: 'Installing, this may take a few minutes...', 'Please create a default UNIX user account. The username does not need to match your Windows username.', 'For more information visit: https://aka.ms/wslusers', and 'Enter new UNIX username: ' followed by a cursor. A vertical scrollbar is visible on the right side of the terminal.

```
Ubuntu
Installing, this may take a few minutes...
Please create a default UNIX user account. The username does not need to match your Windows username.
For more information visit: https://aka.ms/wslusers
Enter new UNIX username: 
```

Configure Ubuntu

Finally, it's always good practice to install the latest updates with the following commands, entering your password when prompted.

`sudo apt update`

```
vlukaj@DESKTOP-P2RPBKF:~$ sudo apt update
[sudo] password for vlukaj:
Hit:1 http://archive.ubuntu.com/ubuntu bionic InRelease
Get:2 http://archive.ubuntu.com/ubuntu bionic-updates InRelease [88.7 kB]
Get:3 http://security.ubuntu.com/ubuntu bionic-security InRelease [88.7 kB]
Get:4 http://archive.ubuntu.com/ubuntu bionic-backports InRelease [83.3 kB]
Get:5 http://archive.ubuntu.com/ubuntu bionic/universe amd64 Packages [8570 kB]
9% [3 InRelease gpgv 88.7 kB] [5 Packages 1923 kB/8570 kB 22%]_
```

Configure Ubuntu

Finally, it's always good practice to install the latest updates with the following commands, entering your password when prompted.

`sudo apt update`

Then

`sudo apt upgrade`

Press Y when prompted.

```
vlukaj@DESKTOP-P2RPBKF: ~  
get:21 http://security.ubuntu.com/ubuntu bionic-security/multiverse Translation-en [3836 B]  
get:22 http://archive.ubuntu.com/ubuntu bionic-updates/universe Translation-en [401 kB]  
get:23 http://archive.ubuntu.com/ubuntu bionic-updates/multiverse amd64 Packages [30.0 kB]  
get:24 http://archive.ubuntu.com/ubuntu bionic-updates/multiverse Translation-en [7532 B]  
get:25 http://archive.ubuntu.com/ubuntu bionic-backports/main amd64 Packages [53.2 kB]  
get:26 http://archive.ubuntu.com/ubuntu bionic-backports/main Translation-en [14.5 kB]  
get:27 http://archive.ubuntu.com/ubuntu bionic-backports/universe amd64 Packages [18.1 kB]  
get:28 http://archive.ubuntu.com/ubuntu bionic-backports/universe Translation-en [8668 B]  
Fetched 26.4 MB in 8s (3401 kB/s)  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
58 packages can be upgraded. Run 'apt list --upgradable' to see them.  
vlukaj@DESKTOP-P2RPBKF:~$ sudo apt upgrade  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
Calculating upgrade... Done  
The following package was automatically installed and is no longer required:  
  libdumbnet1  
Use 'sudo apt autoremove' to remove it.  
The following NEW packages will be installed:  
  dbus-user-session dbus-x11 distro-info fontconfig-config fonts-dejavu-core libasynclns0 libdrm-amdgpu1 libdrm-intel1  
  libdrm-nouveau2 libdrm-radeon1 libflac8 libfontconfig1 libfontenc1 libgl1 libgl1-mesa-dri libgl1-mesa-glx  
  libglapi-mesa libglvnd0 libglx-mesa0 libglx0 libice6 liblvm10 libnetplan0 libogg0 libpciaccess0 libpulse0  
  libpulsedsp libsensors4 libsm6 libsndfile1 libvorbis0a libvorbisenc2 libx11-xcb1 libxaw7 libxcb-dri2-0 libxcb-dri3-0  
  libxcb-glx0 libxcb-present0 libxcb-shape0 libxcb-sync1 libxcomposite1 libxdamage1 libxfixes3 libxft2 libxi6  
  libxinerama1 libxmu6 libxpm4 libxrandr2 libxrender1 libxshmfence1 libxt6 libxtst6 libxv1 libxxf86dga1 libxxf86vm1  
  motd-news-config pulseaudio-utils python3-pexpect python3-ptyprocess x11-common x11-utils  
The following packages will be upgraded:
```

Installate your first package

Installing packages on Ubuntu is as easy as using a single command. Below, you will see how to install `bpython`, a simple python interpreter for trying out ideas, featuring some nice usability features like expected parameters and autocompletion.

Installate your first package

To check that you have the latest package lists, type:

```
sudo apt update
```

Then install bpython:

```
sudo apt install bpython
```

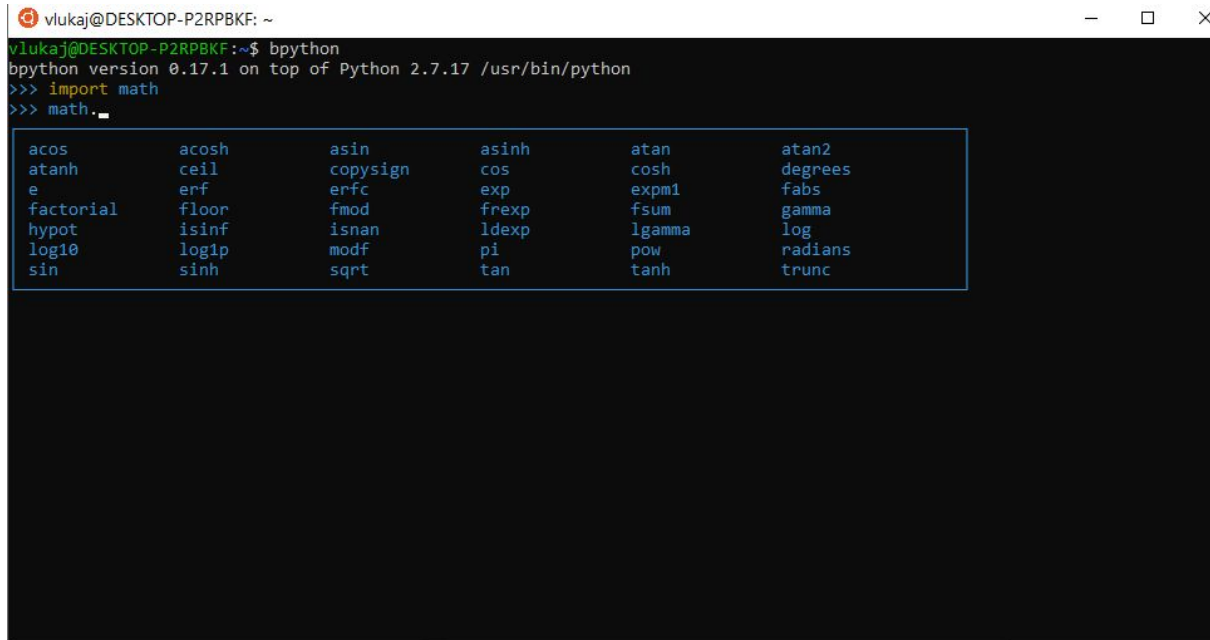
```
vlukaj@DESKTOP-P2RPBKF: ~  
Running hooks in /etc/ca-certificates/update.d...  
done.  
vlukaj@DESKTOP-P2RPBKF:~$ sudo apt install bpython  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
The following package was automatically installed and is no longer required:  
  libdumbnet1  
Use 'sudo apt autoremove' to remove it.  
The following additional packages will be installed:  
  javascript-common libjs-jquery libjs-sphinxdoc libjs-underscore libpython-stdlib libpython2.7-minimal  
  libpython2.7-stdlib python python-argcomplete python-argh python-asn1crypto python-blessings python-certifi  
  python-cffi-backend python-chardet python-cryptography python-curtsies python-enum34 python-greenlet python-idna  
  python-ipaddress python-minimal python-openssl python-pathtools python-pkg-resources python-pygments python-requests  
  python-six python-urllib3 python-urwid python-watchdog python-wcwidth python-yaml python2.7 python2.7-minimal  
Suggested packages:  
  python-jedi python-twisted apache2 | lighttpd | httpd python-doc python-tk python-cryptography-doc  
  python-cryptography-vectors python-enum34-doc python-greenlet-doc python-greenlet-dev python-greenlet-dbg  
  python-openssl-doc python-openssl-dbg python-setuptools ttf-bitstream-vera python-socks python-ntlm python-urwid-doc  
  python2.7-doc binutils binfmt-support  
The following NEW packages will be installed:  
  bpython javascript-common libjs-jquery libjs-sphinxdoc libjs-underscore libpython-stdlib libpython2.7-minimal  
  libpython2.7-stdlib python python-argcomplete python-argh python-asn1crypto python-blessings python-certifi  
  python-cffi-backend python-chardet python-cryptography python-curtsies python-enum34 python-greenlet python-idna  
  python-ipaddress python-minimal python-openssl python-pathtools python-pkg-resources python-pygments python-requests  
  python-six python-urllib3 python-urwid python-watchdog python-wcwidth python-yaml python2.7 python2.7-minimal  
0 upgraded, 36 newly installed, 0 to remove and 0 not upgraded.  
Need to get 6537 kB of archives.  
After this operation, 30.0 MB of additional disk space will be used.  
Do you want to continue? [Y/n]
```

Installate your first package

To run the application type:

bpython

And you're ready to go!



A terminal window titled 'vlukaj@DESKTOP-P2RPBKF: ~' showing the execution of 'bpython'. The output indicates 'bpython version 0.17.1 on top of Python 2.7.17 /usr/bin/python'. The user enters '>>> import math' and '>>> math.', which triggers a display of the math module's attributes. These attributes are listed in a grid-like format within a light blue box. The attributes include: acos, acosh, asin, asinh, atan, atan2, atanh, ceil, copysign, cos, cosh, degrees, e, erf, erfc, exp, expm1, fabs, factorial, floor, fmod, frexp, fsum, gamma, hypot, isinf, isnan, ldexp, lgamma, log, log10, log1p, modf, pi, pow, radians, sin, sinh, sqrt, tan, tanh, and trunc.

```
vlukaj@DESKTOP-P2RPBKF: ~  
vlukaj@DESKTOP-P2RPBKF:~$ bpython  
bpython version 0.17.1 on top of Python 2.7.17 /usr/bin/python  
>>> import math  
>>> math.  
acos      acosh     asin      asinh     atan      atan2  
atanh     ceil      copysign  cos       cosh      degrees  
e          erf       erfc      exp       expm1     fabs  
factorial floor     fmod      frexp     fsum      gamma  
hypot     isinf    isnan     ldexp     lgamma    log  
log10     log1p    modf      pi        pow       radians  
sin       sinh     sqrt      tan       tanh      trunc
```

Customising your terminal with Windows Terminal Preview

Install Windows Terminal Preview to get the most user-friendly setup. You can find it in the [Microsoft Store](#).

Microsoft Store

Cerca app, giochi, filmati e altro ancora

App
Gioco
Film e TV

Windows Terminal Preview
Microsoft Corporation

Ottieni

4,9 ★ Media 9 Classificazioni

Questa è una versione di anteprima del Terminale Windows, che contiene le funzionalità più recenti man mano che...

Strumenti di sviluppo

3 PEGI 3 Fascia d'età: PEGI, PEGI 3

Raccolta Guida

Screenshot

```
PowerShell 7.0.0
Copyright (c) Microsoft Corporation. All rights reserved.
https://aka.ms/powershell
Type 'help' to get help.

Loading personal and system profiles took 61ms.
c:\Users\matt\OneDrive\Documents> cd ~/GitHub/WindowsTerminal
c:\Users\matt\OneDrive\Documents> master 12 +5 -0 -0 1
Running:
Pre-requisites ..... 3/3 Pass
Validate Data ..... 8/8 Pass
Process Data ..... 0/0 Pass
Generate Results ..... 2/3 Partial
Validate Results ..... 1/3 Fail
Results:
3/3 Pass
2/3 Partial
1/3 Fail

c:\Users\matt\OneDrive\Documents> master 12 +5 -0 -0 1
```

Descrizione

Questa è una versione di anteprima del Terminale Windows, che contiene le funzionalità più recenti man mano che vengono sviluppate. Terminale Windows è un'applicazione terminale moderna, veloce, efficiente, utile e produttiva per gli utenti che utilizzano shell e strumenti da riga di comando come il prompt dei comandi, PowerShell e WSL. Le sue funzionalità principali includono più schede, riquadri, supporto dei caratteri Unicode e UTF-8, un motore di rendering del testo con accelerazione GPU, oltre a configurazioni, stili e temi personalizzati.

Si tratta di un progetto open source in cui la partecipazione della community è ben gradita. Per partecipare, visita la pagina <https://github.com/microsoft/terminale>

Recensioni e valutazioni

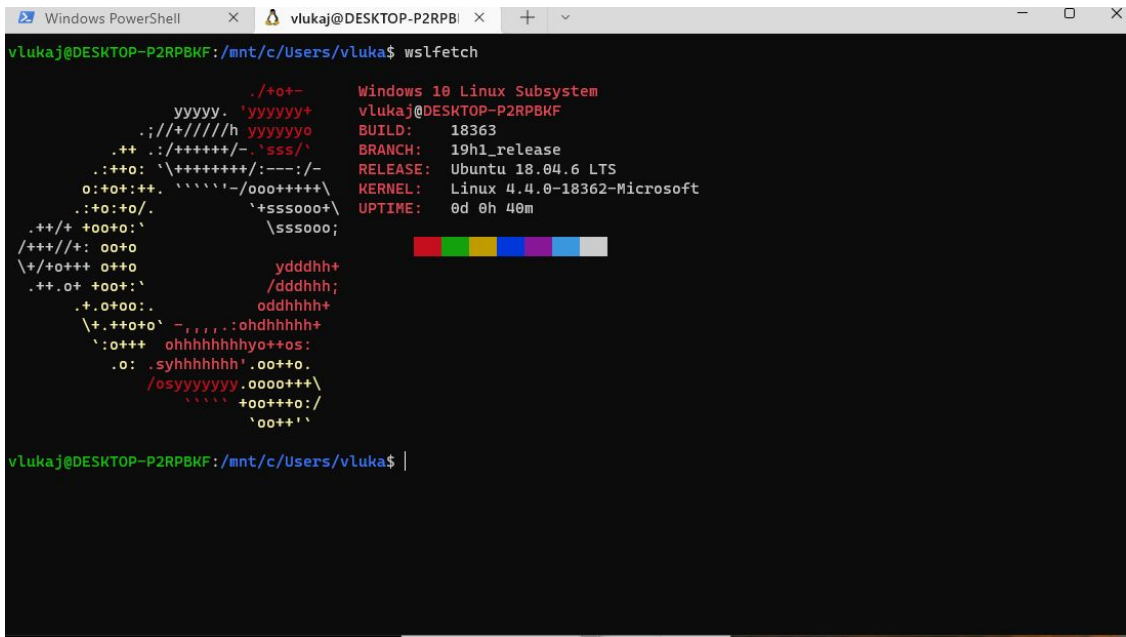
4.9

5 ★
4 ★
3 ★

Customising your terminal with Windows Terminal Preview

Windows Terminal allows you to open multiple Terminal instances as tabs, so you can have PowerShell running alongside Ubuntu. It also includes a number of customisation options. In the below screenshot, we've changed the tab name and colour, and configured the terminal appearance to use the Tango Dark theme and the Ubuntu font!

wslfetch



```
Windows PowerShell x vlukaj@DESKTOP-P2RPBFI x + v
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka$ wslfetch

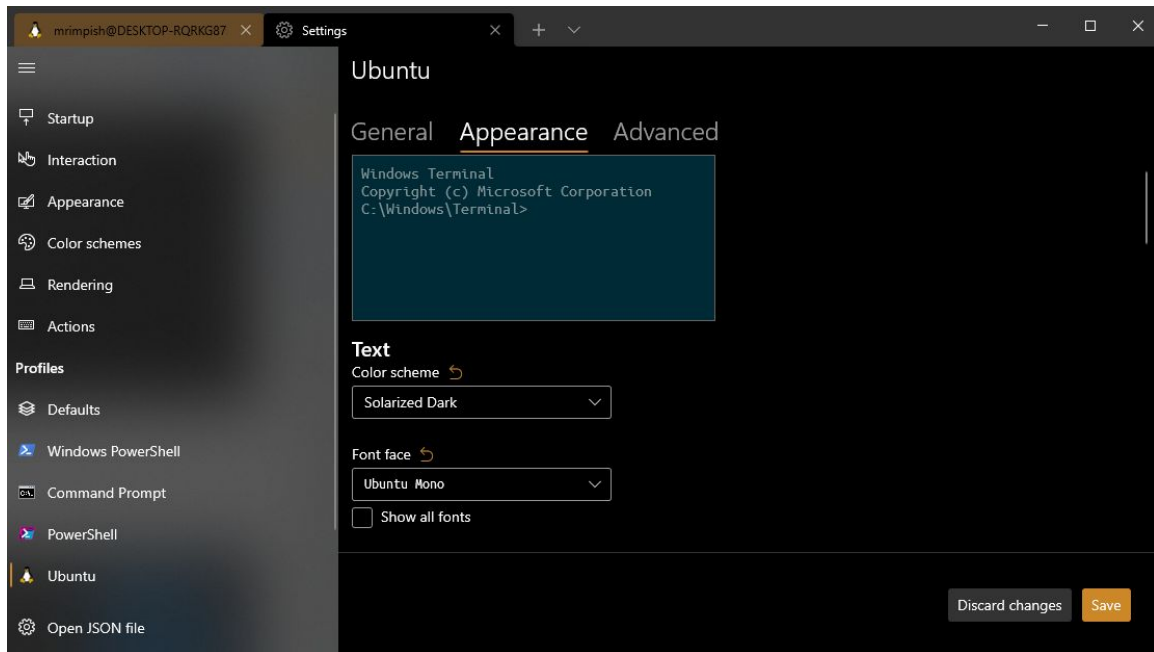
                                ./+o+-
                                yyyyy. 'yyyyyy+
                                .; //+////h yyyyyyo
                                .+ .:/+++++/-.'sss/'
                                .:++o: '\+++++++/:---:/-
                                o:+o+:++. '-----/ooo+++++\
                                .:o:+o/.' 'sSSooo+\
                                .++/+ +oo+o:' \SSSooo;
                                /+++//+: oo+o
                                \+/+o+++ o++o
                                .++ .o+ +oo+: '
                                .+.o+oo:.
                                \+.++o+o' -.,.,.,:ohdhhhhh+
                                ':o+++ ohhhhhhhhyo++os:
                                .o: .syhhhhhhh'.oo++o.
                                /osyyyyyy.oooo+++\
                                +oooo+:/
                                'oo++'

Windows 10 Linux Subsystem
vlukaj@DESKTOP-P2RPBKF
BUILD: 18363
BRANCH: 19h1_release
RELEASE: Ubuntu 18.04.6 LTS
KERNEL: Linux 4.4.0-18362-Microsoft
UPTIME: 0d 0h 40m

vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka$ |
```

Customising your terminal with Windows Terminal Preview

These customisations can be applied universally using the Appearance menu in Settings or to individual profiles which each have their own Appearance menu. Try it yourself to find something you feel comfortable with!



Run First Command

| “~” is the
location of your
home directory

```
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka$ pwd
/mnt/c/Users/vluka
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka$ cd ~
vlukaj@DESKTOP-P2RPBKF:~$ |
```

Run First Command

|“..” is the
location of the
directory below
current one

```
vlukaj@DESKTOP-P2RPBKF:/home$ cd ..  
vlukaj@DESKTOP-P2RPBKF:/$ |
```

Run First Command

To list the files
in the current
directory use “ls”

```
vlukaj@DESKTOP-P2RPBKF:/$ ls
bin boot dev etc home init lib lib64 media mnt opt proc root run
vlukaj@DESKTOP-P2RPBKF:/$
```

Command ls

- **ls** has many options
 - **-l** long list (displays lots of info)
 - **-t** sort by modification time
 - **-S** sort by size
 - **-h** list file sizes in human readable format
 - **-r** reverse the order
- “**man ls**” for more options
- Options can be combined: “**ls -ltr**”

Command ls -ltr

List files by time in reverse order with long listing

```
lukaj@DESKTOP-P2RPBKF:/$ ls -ltr
total 620
drwxr-xr-x  1 root root   4096 Mar 21  2019 snap
drwxr-xr-x  1 root root   4096 May 21  2019 srv
drwxr-xr-x  1 root root   4096 May 21  2019 opt
drwxr-xr-x  1 root root   4096 May 21  2019 media
drwxr-xr-x  1 root root   4096 May 21  2019 usr
drwx----- 1 root root   4096 May 21  2019 root
drwxr-xr-x  1 root root   4096 May 21  2019 var
drwxr-xr-x  1 root root   4096 May 21  2019 boot
-rwxr-xr-x  1 root root 632048 Apr  5  2021 init
dr-xr-xr-x 12 root root     0 Nov  9 11:06 sys
dr-xr-xr-x 12 root root     0 Nov  9 11:06 proc
drwxr-xr-x  1 root root   4096 Nov  9 11:06 mnt
drwxr-xr-x  1 root root   4096 Nov  9 11:07 home
drwxr-xr-x  1 root root   4096 Nov  9 11:21 lib64
drwxr-xr-x  1 root root   4096 Nov  9 11:22 lib
drwxr-xr-x  1 root root   4096 Nov  9 11:23 bin
drwxr-xr-x  1 root root   4096 Nov  9 11:23 sbin
drwxr-xr-x  1 root root   4096 Nov  9 11:24 run
drwxr-xr-x  1 root root   4096 Nov  9 11:31 etc
drwxrwxrwt  1 root root   4096 Nov  9 11:32 tmp
drwxr-xr-x  1 root root   4096 Nov  9 11:46 dev
lukaj@DESKTOP-P2RPBKF:/$
```

Command: mkdir

To create a new directory use “mkdir”

```
/lukaj@DESKTOP-P2RPBKF:/$ sudo mkdir test  
[sudo] password for vlukaj:  
/lukaj@DESKTOP-P2RPBKF:/$ |
```

```
vlukaj@DESKTOP-P2RPBKF:/$ ls  
bin  dev  home  lib    media  opt    root  sbin  srv  test  usr  
boot  etc  init  lib64  mnt    proc   run   snap  sys  tmp   var  
vlukaj@DESKTOP-P2RPBKF:/$ |
```

Command: rmdir

To remove an empty directory use
“rmdir”

```
vlukaj@DESKTOP-P2RPBKF:/$ sudo rmdir test
vlukaj@DESKTOP-P2RPBKF:/$ ls
bin  boot  dev  etc  home  init  lib  lib64  media  mnt  opt  proc  root  run  sbin  snap  srv  sys  tmp  usr  var
vlukaj@DESKTOP-P2RPBKF:/$ |
```

Create a file: nano

```
vlukaj@DESKTOP-P2RPBKF:/test$ sudo nano test
vlukaj@DESKTOP-P2RPBKF:/test$ ls
test
vlukaj@DESKTOP-P2RPBKF:/test$ |
```



The screenshot shows the GNU nano 2.9.3 text editor interface. The title bar at the top indicates the file being edited is 'test'. The main editing area contains the text 'ciao a tutti' on the first line. At the bottom, a status bar displays various keyboard shortcuts for navigation and editing. A message '[File 'test' is unwritable]' is visible in the status bar, indicating that the file is read-only.

```
GNU nano 2.9.3 test
ciao a tutti

[ File 'test' is unwritable ]
^G Get Help  ^O Write Out  ^W Where Is  ^K Cut Text  ^J Justify   ^C Cur Pos  ^U Undo      ^M Mark Text
^X Exit      ^R Read File  ^B Replace  ^N Uncut Text ^_ To Spell  ^G Go To Line ^F Redo     ^D Copy Text
```